

Practical 6

17ce049

Visualizing the Titanic Disaster ¶

Introduction:

This exercise is based on the titanic Disaster dataset available at [Kaggle \(https://www.kaggle.com/c/titanic\)](https://www.kaggle.com/c/titanic). To know more about the variables check [here \(https://www.kaggle.com/c/titanic/data\)](https://www.kaggle.com/c/titanic/data).

Step 1. Import the necessary libraries

In [2]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

%matplotlib inline
```

Step 2. Import the dataset from this [address](https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization)
https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization

Step 3. Assign it to a variable titanic

In [4]:

```
url = 'https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/Visualization/Titanic.csv'

titanic = pd.read_csv(url)

titanic.head()
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

Step 4. Set PassengerId as the index

In [5]:

```
titanic.set_index('PassengerId').head()
```

Out[5]:

	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
PassengerId										
1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C101
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C103
5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN



Step 5. Create a pie chart presenting the male/female proportion

In [24]:

```
# sum the instances of males and females
males = (titanic['Sex'] == 'male').sum()
females = (titanic['Sex'] == 'female').sum()

# put them into a list called proportions
proportions = [males, females]

# Create a pie chart
plt.pie(
    # using proportions
    proportions,

    # with the labels being officer names
    labels = ['Males', 'Females'],

    # with no shadows
    shadow = False,

    # with colors
    colors = ['blue', 'red'],

    # with one slide exploded out
    explode = (0.15, 0),

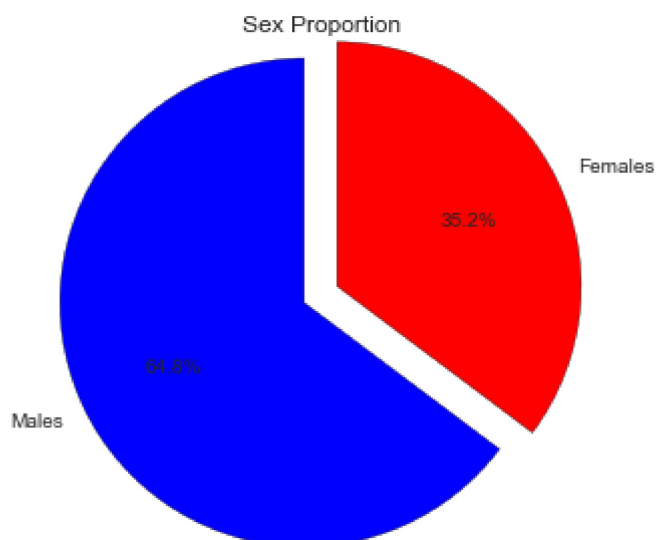
    # with the start angle at 90%
    startangle = 90,

    # with the percent listed as a fraction
    autopct = '%1.1f%%'
)

# View the plot drop above
plt.axis('equal')

# Set Labels
plt.title("Sex Proportion")

# View the plot
plt.tight_layout()
plt.show()
```



Step 6. Create a scatterplot with the Fare payed and the Age, differ the plot color by gender

In [67]:

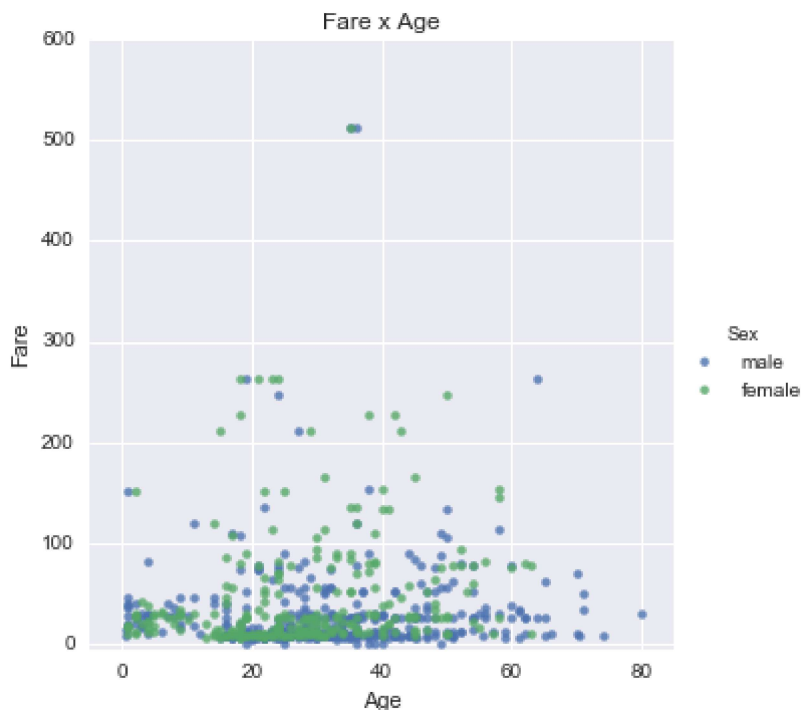
```
# creates the plot using
lm = sns.lmplot(x = 'Age', y = 'Fare', data = titanic, hue = 'Sex', fit_reg=False)

# set title
lm.set(title = 'Fare x Age')

# get the axes object and tweak it
axes = lm.axes
axes[0,0].set_ylim(-5,)
axes[0,0].set_xlim(-5,85)
```

Out[67]:

(-5, 85)



Step 7. How many people survived?

In [68]:

```
titanic.Survived.sum()
```

Out[68]:

342

Step 8. Create a histogram with the Fare payed

In [48]:

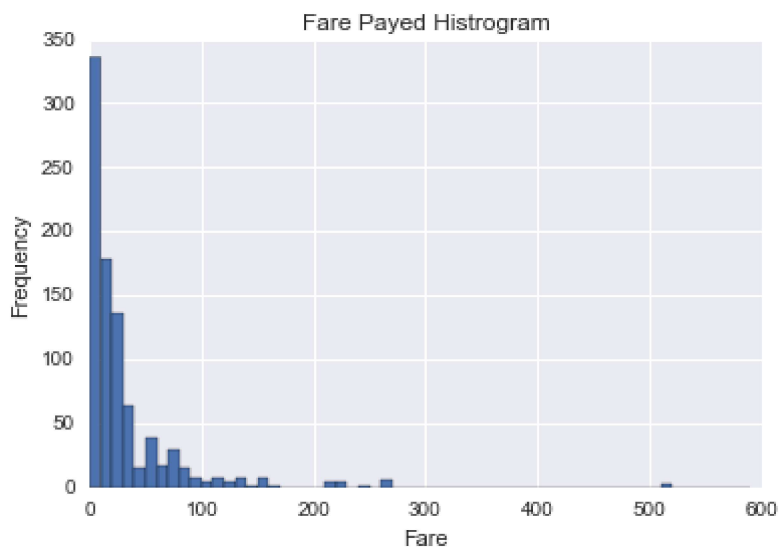
```
# sort the values from the top to the least value and slice the first 5 items
df = titanic.Fare.sort_values(ascending = False)
df

# create bins interval using numpy
binsVal = np.arange(0,600,10)
binsVal

# create the plot
plt.hist(df, bins = binsVal)

# Set the title and labels
plt.xlabel('Fare')
plt.ylabel('Frequency')
plt.title('Fare Payed Histogram')

# show the plot
plt.show()
```



BONUS: Create your own question and answer it.

In []: